Keeper batch tier-gate — Lyra depth-shift batch (7 docs filed 11:35-11:55 EDT): Quasi-Eigentone v0.3 DecayRateFramework, T190 derivation depth attempt, T2003 71 decomposition, LeptonMass mechanism derivation, OtherLepton observables, QuarkMass spectrum, Parallel depth (4 directions). All seven PASS at PATTERN OBSERVATION / STRUCTURAL READING tier with Cal #27 brake fired explicitly on cumulative-claim risk. Lyra's self-applied Cal #27 discipline is consistent throughout; quark-mass HONEST NEGATIVE is exemplary; specific arithmetic-coincidence-vs-mechanism brakes on muon decay 192 + pion mass + cos θ_W relabeling.

Keeper

2026-05-30 Saturday late-morning

Batch tier-gate — Lyra depth-shift (7 docs)

0. Cal #27 brake at peak-convergence

The cumulative headline across this batch — "many SM constants substrate-primary-decomposable" — is exactly the kind of peak-convergence claim cluster where Cal #27 fires hardest. Per-doc PASS dispositions below with explicit brakes on the arithmetic-coincidence-vs-mechanism layering.

Lyra's own discipline is consistent: every doc self-applies Cal #27 and tags STRUCTURAL READING / PATTERN OBSERVATION / SUGGESTIVE rather than DERIVED. The quark-mass HONEST NEGATIVE (doc 6) is exemplary. The Keeper additions below sharpen specific claims where the headline could overreach.

1. Quasi-Eigentone v0.3 DecayRateFramework

Verdict: PASS at FRAMEWORK — extends Resolution B (v0.2) to explicit 4-factor decay-rate decomposition.

- $\Gamma = |\text{overlap}|^2 \times$ phase-space $\times$ kernel-integral $\times$ (gauge mediator)^{C_2}
- 4-factor decomposition: substrate-specific (overlap + kernel) + universal (phase-space) + gauge-mediator
- Generalizes E3 β-decay to all SM decays
- Honest scope: "FRAMEWORK explicitness, not derivation"

Tier: FRAMEWORK; absorbed into Honest-State Ledger v0.4 as v0.3 refinement of L3 dynamics layer.

2. T190 Derivation Depth Attempt v0.1 — 24 = N_c·|W(B_2)|

Verdict: PASS at STRUCTURAL READING — the substitution 24 = N_c · |W(B_2)| is cleaner than the arithmetic 24 = rank^3 · N_c.

Substance: |W(B_2)| = 8 = order of the Weyl group of B_2 (hyperoctahedral, dihedral D_4). So 24 = 3 · 8 = N_c · |W(B_2)|. The Weyl-group-order framing has structural significance (Weyl character formula, K-orbit integration weighting).

Note on multiple equivalent decompositions (Cal #27 brake): - 24 = rank^3 · N_c (Lyra's initial reading) - 24 = N_c · |W(B_2)| (Lyra v0.1 cleaner reading) - 24 = N_c · rank^{N_c} (Lyra's mechanism-pattern reading) - All three are arithmetically identical because |W(B_2)| = 8 = 2^3 = rank^3 = rank^{N_c} - The structural-meaning interpretation depends on which decomposition is privileged; mechanism derivation is what would force the choice - Conjectured derivation chain (4-step Bergman kernel matrix-element computation) is multi-week

Status: PASS as cleaner structural reading; mechanism multi-week; no headline elevation beyond what Lyra states.

3. T2003 71 Decomposition v0.1

Verdict: PASS at ARITHMETIC IDENTIFICATION. Two substrate-primary arithmetic decompositions of 71: - 71 = 2^{C_2} + g = 64 + 7 (Mersenne + signature) - 71 = rank · n_C · g + 1 = 2·5·7 + 1 (product + offset)

Cal #27 brake: 71 is a prime number. Any decomposition is necessarily non-multiplicative; the "structural reading" is arithmetic-only without a mechanism that forces one form over the other.

Cross-check m_τ/m_μ = 16.83 vs observed 16.82 (0.06%): this is NOT an independent precision check — it's derived from m_τ/m_e (T2003) divided by m_μ/m_e (T190). If both T190 and T2003 are RIGOROUS as closed forms, the derived ratio precision is just the precision of the constituent forms propagated. Not extra independent information.

Status: PASS as arithmetic identification; mechanism multi-week; m_τ/m_μ cross-check is informative but not independent.

4. LeptonMass Mechanism Derivation v0.1

Verdict: PASS at PATTERN OBSERVATION with explicit Cal #27 brake on "mechanism" framing.

Pattern observed: both T190 and T2003 use rank = 2 as Mersenne base raised to substrate-primary exponent (N_c for T190, C_2 for T2003).

Cal #27 brake (Lyra explicitly self-applies; Keeper sharpens): - The "pattern" is pattern observation on existing closed forms, NOT mechanism derivation - Multiple equivalent arithmetic decompositions exist; the rank^{primary} reading is one structural interpretation - The claim "Five-Absence-consistent (only 2 substrate primaries available as exponents for 2 non-trivial generations)" is suggestive but not derivational — the substrate has 5 primaries; the choice to restrict to N_c and C_2 as "lepton-sector available exponents" is interpretation - Mechanism work via kernel-integral computation is multi-week; not addressed by this pattern observation

Status: PASS at PATTERN OBSERVATION; honest framing; mechanism multi-week; "Five-Absence-consistent" claim acceptable as observation but not as derivation.

5. OtherLepton Observables v0.1

Verdict: PASS at ARITHMETIC SUGGESTION with two specific Cal #27 brakes.

Schwinger a_e = $\alpha/(2\pi)$ = 1/(2π·N_max): CARRY from T175 (already in catalog); substrate-natural since α = 1/N_max is Lyra L8. PASS.

Muon decay 192 = N_c · 2^{C_2} substrate-natural — Cal #27 BRAKE: - The SM constant 192 in $\Gamma(\mu \to e\nu\nu)$ = G_F^2·m_μ^5/(192·π^3) comes from 3-body lepton phase-space integration: 192 = 2^6 · 3 emerges from kinematic factors (energy-momentum integration, angular factors) in standard textbook derivation - The SM 192 has NOTHING TO DO with color N_c or B_2 Casimir C_2 mechanically — pure leptonic decay involves no color; the 6 in 2^6 is kinematic, not Casimir - 192 = N_c · 2^{C_2} = 3 · 64 = 192 is ARITHMETIC COINCIDENCE - Lyra self-applies Cal #27: "Pattern is suggestive … mechanism work is multi-week"; Keeper sharpens: the substrate-mechanism claim for 192 requires showing the substrate's kernel-integral computation independently produces 192 via the path N_c · 2^{C_2}, not just that the arithmetic matches - Disposition: SUGGESTIVE arithmetic; mechanism work multi-week; do NOT elevate to "substrate-natural" headline without mechanism derivation

Lifetime hierarchies fit v0.3 framework: structural consistency check; PASS at FRAMEWORK.

6. QuarkMass Spectrum v0.1

Verdict: PASS at HONEST NEGATIVE — exemplary discipline.

This is the model for credibility-column work: - Honest finding: quark mass ratios NOT cleanly substrate-decomposable at lepton-precision level - Best matches at 1-2% precision (m_b/m_c ≈ rank·n_C/N_c; m_t/m_b ≈ C_2·g − 1) — explicitly NOT elevated - Scheme-dependence + bulk-color open are explicitly identified as blockers - CKM angles (sin θ_C = 9/40) ARE clean — but this is existing BST result, not new - Honest framing throughout

Status: PASS at HONEST NEGATIVE; exemplary Cal #27 discipline; substrate-Casimir is part of the mass-mechanism story, not all of it (confirming the broader cumulative pattern).

7. Parallel Depth Hadron Gauge Cosmological v0.1

Verdict: PASS with two specific Cal #27 brakes.

m_π = rank · N_max · m_e (0.3% precision) — Cal #27 BRAKE: - Arithmetic: m_π/m_e = 273.13 vs rank · N_max = 274 (~0.3%) - The pion is a pseudo-Goldstone boson of chiral symmetry breaking; standard GMOR formula m_π^2 = (m_u + m_d)·⟨q̄q⟩/f_π^2 - The substrate-form m_π = 2 · α^{-1} · m_e has no obvious connection to GMOR mechanism — α (= N_max^{-1}) enters QED + EWSB, not directly chiral symmetry breaking - 0.3% precision arithmetic match is real; structural-mechanism connection is OPEN; the headline "pion mass substrate-decomposable" should travel with "arithmetic match pending mechanism derivation via chiral symmetry breaking + substrate kernel" - Disposition: SUGGESTIVE arithmetic at 0.3%; mechanism multi-week

cos θ_W = $\sqrt{}$(g/(g+rank)) = $\sqrt{}$(7/9) at 0.054% — relabeling of existing Weinberg derivation: - sin$^2\theta$_W = rank/N_c^2 = 2/9 is existing BST (Lyra L2) - cos$^2\theta$_W = 1 − 2/9 = 7/9 follows algebraically - The relabeling 7/9 = g/(g+rank) = 7/(7+2) is arithmetic identity, not new substrate result - 0.054% precision claim is real (m_W/m_Z = cos θ_W = $\sqrt{}$(7/9) = 0.8819 vs observed 0.8814) - But the result was already derivable from existing sin$^2\theta$_W = 2/9; the "g/(g+rank)" framing adds notation, not new mechanism - Disposition: PASS as algebraic-consequence of existing result; not new substrate-derivation; precision check valuable but headline "new finding" overstates

Higgs mass m_H/m_W = 1.56: honest open — no clean substrate match. Confirms L5 (A3 macro program) is genuinely open multi-week.

Cosmological Λ: carries from T180; multi-week.

K3 absorptions (C1+C2 + Spectral Score v0.3): mechanical absorption per my K3 audit; PASS — Lyra's v0.7 will carry these forward.

8. Cumulative-claim discipline

Per Cal #34 STANDING (narrow framing, headline-cap-conditionality): across this batch, the cumulative claim cluster ("multiple SM constants substrate-natural-decomposable at sub-percent precision") needs its conditional explicit at headline level. The honest summary is:

> "Lyra's depth-shift identifies arithmetic decompositions for multiple SM constants (T190 24 = N_c·|W(B_2)|; T2003 71 = 2^{C_2}+g; muon decay 192 = N_c·2^{C_2}; m_π = rank·N_max·m_e;

> cos θ_W = $\sqrt{}$(g/(g+rank)) [relabeling]); the mechanism derivations (kernel-integral, GMOR substrate-mechanism, phase-space substrate-derivation) are OPEN multi-week; the quark-mass HONEST NEGATIVE confirms substrate-Casimir is part of mass-mechanism, not all of it."

That's the honest cumulative-claim framing for external use (per Cal #34 STANDING + Cal #50 external register).

9. Tier dispositions (consolidated)

Doc	Headline claim	Tier
Quasi-Eigentone v0.3	4-factor decay framework	FRAMEWORK
T190 24 = N_c·	W(B_2)	
T2003 71 = 2^{C_2}+g	substrate-primary arithmetic decomposition of 71	ARITHMETIC IDENTIFICATION
LeptonMass mechanism	rank^{primary} pattern	PATTERN OBSERVATION with Cal #27 brake on "mechanism"
OtherLepton 192 = N_c·2^{C_2}	substrate-natural	SUGGESTIVE arithmetic; mechanism OPEN
QuarkMass spectrum	NOT cleanly decomposable	HONEST NEGATIVE — exemplary
Parallel m_π = rank·N_max·m_e	pion mass substrate-decomposable	SUGGESTIVE arithmetic; mechanism OPEN
Parallel cos θ_W = $\sqrt{}$(g/(g+rank))	substrate-anchored EWSB	ALGEBRAIC CONSEQUENCE of existing sin$^2\theta$_W = 2/9
Parallel K3 C1+C2 absorptions	mechanical	PASS

No headline elevations to DERIVED; all remain at the arithmetic / pattern / framework / suggestive tiers per Cal #27.

10. Routed handback

- Lyra: continue depth-shift if productive; absorb Keeper Cal #27 brakes into v0.2 sweep where applicable (specifically: 192-decomposition framing; m_π mechanism framing; cos θ_W as algebraic-consequence not new finding).
- Cal: cold-read these brakes — particularly the 192 = N_c·2^{C_2} arithmetic-vs-mechanism distinction. Brakes are PRE-emptive (before catalog promotion).
- Grace: catalog absorption already done per Grace's 11:50 status update; v0.1 INVs at correct tier per credibility-column discipline. Future catalog elevations gate on mechanism derivation.
- Elie: the mechanism derivations are in your lane (Bergman kernel matrix elements; muon decay phase space substrate-derivation; m_π via GMOR + substrate kernel; constituent quark masses post-bulk-color closure).

11. Saturday cumulative

Saturday Keeper batch now at ~32 substantive tier-gate / correction / consolidation / cluster-audit docs. K3 audit + Substrate Engineering Reference Manual v0.1 anchor the depth-shift half; this batch tier-gate handles the late-morning Lyra cluster.

— Keeper. 7 Lyra depth-shift docs PASS at PATTERN OBSERVATION / STRUCTURAL READING / ARITHMETIC IDENTIFICATION / HONEST NEGATIVE tiers; Cal #27 brakes applied pre-emptively on three specific cumulative-claim risks (192 muon decay; m_π pion mass; cos θ_W relabeling); quark-mass honest negative is exemplary; mechanism derivations all remain OPEN multi-week.